

Azure Pipeline Troubleshooting Real-World Cheat Sheet

A curated list of most useful and commonly used **commands and techniques** to troubleshoot Azure Pipeline failures in **real-time projects**.

1. Check Pipeline Run Status

```
az pipelines runs list --org https://dev.azure.com/<org> --project <project-name> \
  --query "[].{ID:id,Status:status,Result:result}" -o table
```

Quickly see recent pipeline run statuses.

2. Get Detailed Logs of a Failed Run

```
az pipelines runs show --id <run-id> --org https://dev.azure.com/<org> --project <project-name> --output json
```

Grab error messages, logs, and stages for deep failure analysis.

3. Download Full Logs

```
az pipelines runs artifact download --run-id <run-id> --artifact-name 'logs' --path ./logs
```

Useful for offline debugging or sharing with team.

4. Rerun a Failed Pipeline

```
az pipelines run --name <pipeline-name> --branch <branch-name> \
  --org https://dev.azure.com/<org> --project <project-name>
```

Re-run the same pipeline after fixing an issue.

5. Check Default Branch

```
az pipelines show --name <pipeline-name> --query  
"repository.defaultBranch"
```

Diagnose misconfigured triggers or branch issues.

6. Echo Environment Variables (In YAML or Bash)

```
echo $BUILD_REASON  
echo $BUILD_SOURCEBRANCH  
echo $SYSTEM_DEBUG
```

Helps validate trigger type, source branch, and debug flag.

7. Enable Command Tracing in Shell Scripts

```
#!/bin/bash  
set -x # Echo all commands for debugging
```

Logs each command for easier tracing.

8. Troubleshoot Docker Task Failures

```
docker logs <container-id>  
docker ps -a
```

Find container errors in Docker-based pipeline stages.

9. Validate YAML Locally

```
az pipelines validate --name <pipeline-name> --path azure-  
pipelines.yml --project <project-name>
```

Avoid breaking your pipeline with bad YAML.

10. Check Agent Pool/Status

```
az pipelines agent list --pool-name Default --project  
<project-name>  
az pipelines agent show --agent-id <id> --pool-name Default  
--project <project-name>
```

Check if agents are offline, stuck, or misconfigured.

11. Verify Service Connections

```
az devops service-endpoint list --project <project-name> --  
org https://dev.azure.com/<org> -o table
```

Detect broken service connections to Azure, GitHub, Docker, etc.

12. Test Variable/Secret Availability

```
- script: |  
    echo "My variable: $(myVariable)"
```

Debug missing or misconfigured variables/secrets.

13. Git Troubleshooting for Pipeline YAML Changes

```
git log -p azure-pipelines.yml  
git blame azure-pipelines.yml
```

See who changed what — and when — in pipeline files.

14. Validate Infra-as-Code (Terraform/ARM)

```
terraform validate  
terraform plan  
az deployment sub validate --location <region> --template-  
file <template.json>
```

Prevent deployment failures before reaching the pipeline.

15. Clean Workspace Between Builds

```
- script: |  
    sudo rm -rf $(Pipeline.Workspace)/*  
    displayName: 'Clean workspace'
```

Fixes errors due to leftover cache/temp files.